

Module 11: Genomics & Biotechnology — Study Questions

1. What is a restriction enzyme? What does it do to DNA?
2. What is a plasmid, and why is it useful in genetic engineering?
3. PCR copies DNA in a test tube. Describe the three basic steps. For each step, name the Module 07 enzyme or concept it is similar to. (Hint: helicase, primase, DNA polymerase.)
4. After 1 cycle of PCR you have 2 copies. After 2 cycles you have 4. How many copies do you have after 5 cycles? After 10?
5. In gel electrophoresis, do small or large DNA fragments travel farther? Why?
6. What are STRs? How are STR patterns inherited? (Connect this to Module 09.)
7. How is DNA fingerprinting used in forensics or paternity testing?
8. What is a GMO? Describe the basic steps to make a bacterium produce a human protein (like insulin).
9. In Module 07 you learned that the genetic code is "universal." Why is this fact absolutely essential for genetic engineering to work?
10. What is CRISPR-Cas9? In simple terms, how does it edit DNA?
11. Name one benefit and one concern about GMOs or gene editing.
12. **Putting it all together:** A child is born with a genetic disorder caused by a single-base mutation (Module 07). The disorder follows an autosomal recessive pattern (Module 09). Scientists want to use CRISPR (Module 11) to fix the gene. In 3–4 sentences, explain how concepts from Modules 07, 09, and 11 all connect in this scenario.